

6.	Ans: B For an ice cube floating on sea water, its weight = upthrust, so $0.920 \text{ g cm}^{-3} \times 1.00 \text{ cm}^3 \times g = 1.03 \text{ g cm}^{-3} \times \text{volume of sea water displaced} \times g$ So volume of sea water displaced = 0.893 cm^3 let the volume of water from melted ice be v, by conservation of mass, we have $0.920 \text{ g cm}^{-3} \times 1.00 \text{ cm}^3 = 1.00 \text{ g cm}^{-3} \times v$ $v = 0.920 \text{ cm}^3$ So volume of liquid increases by $0.920 - 0.893 = 0.03 \text{ cm}^3$
7.	Ans: D